

## KIMBERLY K. BROUGHTON

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Software Engineer with over 17 years of experience designing, developing, and modernizing enterprise software solutions. Proven expertise in Python, JavaScript, automation, APIs, and database design across scientific research, healthcare, publishing, and government environments. Recognized NASA Inventor with a history of delivering scalable applications, reducing technical debt, automating complex workflows, and collaborating directly with stakeholders to transform business requirements into reliable software solutions.

### SIGNIFICANT ACHIEVEMENTS

**Sole architect and lead developer of NASA Earthdata Pub Overview and Forms applications used by all 12 NASA DAACs for all Data Producers. Official NASA "Inventor" of Earthdata Pub (NASA Case No. GSC-18, 612-1).**

- Architecture
- Development
- Cloud migration
- Adoption across NASA DAACs
- APIs
- Dashboard modernization
- Design decisions
- Authentication
- Technical leadership

**Designed and implemented administrative software system to replace data processing department** 02/2018

**UI / UX Certified in User Experience – NN/g Nielsen Norman Group** 12/2020

**Contributed to ONC Stage 1 and Stage 2 Electronic Health Record Certification:** Electronic Health Record Software Certification - 1 of 3 developers assigned

### TECHNICAL SKILLS

**Languages:** Python, JavaScript, SQL, GraphQL, SQLAlchemy, C#, C++, Java, Perl, PHP, HTML5, CSS3, XML, Shell, AWK, Visual Basic, VB.NET

**Frameworks:** Flask, Vue.js, React, Angular, Bootstrap, jQuery, Mustache, Jinja2, Cypress, Webapp2

**Cloud & DevOps:** AWS Cloud, Docker, Kubernetes, Git, GitLab, GitLab CI/CD Pipelines, GitLab Runners, YAML, Automation Scripting

**Databases:** PostgreSQL, Oracle, Microsoft SQL Server, MySQL, Informix, Cache, c-tree, Microsoft Access

**Development Tools:** VS Code, DBVisualizer, Oracle Forms & Reports, Crystal Reports, Jasper Reports, Visual Studio, Swagger, SpectaQL, Sphinx, Vuese

**Content Management:** Drupal, Joomla, WordPress

**AI Development Tools:** Cline, DUO AI, GitHub Copilot

**Operating Systems:** Linux, Microsoft Windows

### CORE COMPETENCIES

- Software Architecture
- Full Stack Development
- Enterprise Applications
- REST APIs
- Database Design
- Automation
- System Integration
- Technical Leadership
- Application Modernization
- Technical Documentation
- User Experience
- Agile Development

### EXPERIENCE

**ORNL DAAC – Oak Ridge, TN**

**04/18 – Present**

*Systems Architect / Application Developer*

Developing in: Python, JavaScript (above frameworks and DevOps), SQLAlchemy, GraphQL, AWS Cloud, Perl, HTML, CSS, SQL, Drupal, DBVisualizer, VSCode

- Repurposed NASA's Cumulus Dashboard into the NASA's Earthdata Pub Dashboard powering my Overview and Forms applications (Vue, React) serving all 12 NASA DAACs and hundreds of data producers
- Co-developed the Earthdata Pub REST API powering multiple NASA applications
- Assisted in refactoring and expansion of River Function Indicator Questionnaire (RFIQ) to a generalized Environmental Decision Support Toolkit
- Designed and implemented MyWorkSpace, a modular data order application expanding user download options
- Was 1 of 3 developers who delivered an Ingest Dashboard for ORNL DAAC ingest operations (MVP live), architected to interface with Earthdata Pub for QA and full automation

- Designed automation scripts that reduced manual processing time by 80%
- Refactored sites and scripts from Perl to Python using an efficient templating system
- Implemented automated unit and end-to-end testing, improving deployment confidence and reducing regression defects
- Implemented metadata import/export pipelines supporting multiple formats and standards
- Modernized and maintained the ORNL DAAC web platform while supporting migration to NASA's cloud-hosted Earthdata environment, improving maintainability and supporting future platform growth

**COSBY HARRISON COMPANY (Formerly TAP Publishing Company) – Crossville, TN**

**10/16 – 04/18**

*System Architect / Web Application Developer / Web Developer*

Developed in: Python, JavaScript, Perl, DOS, AutoHotKey, HTML, CSS, SQL, Oracle, Informix, Shell, C++

- Designed and implemented administrative software system to replace the entire data processing department
- Automated publishing workflows, reducing hours of manual work to a single button click while virtually eliminating errors and improving company costs
- Standardized data across multiple formats via automated import/export pipelines
- Automated tasks via scripts across multiple environments and languages
- Maintained and enhanced custom, publication, and company web platforms with feature and framework improvements

**LIBERA, INC. – Knoxville, TN**

**10/15 – 07/16**

*Web Application Programmer*

Developed in: VB.NET/ASP.NET, XML, C#, TSQL, System7, SQL, Visual Studio 2015

- Implemented Vocational Rehabilitation program updates affecting all agencies statewide
- Created custom queries and scripts enabling agency end users to access timely reporting data
- Performed unit testing for statewide Vocational Rehabilitation program updates supporting developers across the state
- Wrote operating instructions and reference documentation for end users

**RMB, INC. – Knoxville, TN**

**03/15 – 07/15**

*FACS Application Programmer*

Developed in: FACS, VBScript, AutoHotKey, Jasper Reports, Crystal Reports

- Maintained database operations at peak performance, protecting secured information and ensuring data integrity
- Built daily jobs and import/export routines routing high-volume data in and out of production systems
- Created custom queries and reports providing timely and accurate information to end users
- Wrote operating instructions and reference documentation for users

**QRS, INC. – Knoxville, TN**

**05/09 – 02/15**

*EHR Developer / Software Developer*

Developed in: Javascript, HTML, CSS, PHP, Python, Perl, Linux, MySQL, MSSQL, c-tree

- Indirectly led other EHR programmers on client-facing clinical software projects
- Designed Standard Form Sets / Notes tailored to doctor specialty
- Worked directly with doctors to determine practice needs and translate them into software solutions
- Provided Tier 2 product support, resolving escalated client issues
- Created documentation and instructional videos training clients on Meaningful Use compliance and adoption
- Implemented HL7 interfaces and data conversions based on client needs
- Designed back-end and front-end interfaces and custom programs based on client needs

## **EDUCATION**

**UI / UX Certification – NN/g Norman Group**

**12/20**

**C# CERTIFICATION – Knoxville, TN**

**12/17**

**PELLISSIPPI TECHNICAL COMMUNITY COLLEGE – Knoxville, TN**

**05/07 – 05/09**

*A.A.S. Computer Programming*

*A.A.S. Database Design and Development*

- GPA at 3.60
- Dean's List for Fall 2008, Spring 2007